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Personalized Pokémon Generation Using GANs and Deep Feature Extraction

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Abstract — This project uses Generative Adversarial Networks (GANs) combined with deep feature extraction techniques to create personalized Pokémon images based on human visual features. By leveraging a pre-trained VGG16 model, we extract detailed features from both human and Pokémon image datasets. The extracted features are then used to identify the most visually similar Pokémon to a given human image using cosine similarity. A Conditional GAN (CGAN) is used to generate new Pokémon images by blending human-specific attributes with the selected Pokémon, resulting in unique, hybridized characters. The approach integrates convolutional layers for feature learning and GAN-based synthesis, providing an innovative method for image generation that combines both artistic creativity and advanced machine learning techniques. This project tries to show the potential of deep learning for personalized content generation in creative and entertainment applications, offering a novel pipeline for character customization.

Index Terms — VGG16, pre-trained, feature extraction, Generative Adversarial Network(GAN), C-GAN, Pokemon, Human features, Layers, generator, discriminator.

I. INTRODUCTION

Image generation has become quite significant with the rise of deep learning techniques, particularly Generative Adversarial Networks (GANs). These models have shown immense potential in generating realistic images by learning underlying data distributions. In this project, we aim to explore the creative application of GANs for generating personalized Pokémon characters by incorporating visual

features from human images. The goal is to fuse the distinctive attributes of human subjects with the aesthetic elements of Pokémon, resulting in hybrid characters that blend characteristics from both domains.

The core of this project revolves around two key components: deep feature extraction using a pre-trained VGG16 model and image generation using a Conditional GAN (CGAN). The VGG16 model is utilized to extract deep features from human and Pokémon images, capturing essential visual information like facial features, body structure, and textures. Using cosine similarity, we identify the Pokémon that best matches the human image based on their extracted features. The CGAN then generates new Pokémon images by conditioning the generator on both the human features and the identified Pokémon, enabling the creation of unique, personalized characters.

This project not only leverages the power of GANs and deep feature extraction but also demonstrates the potential of machine learning for creative applications, such as character customization and content generation. By combining human and Pokémon attributes, the system provides a novel approach to personalized character design, appealing to fans of the Pokémon franchise and enthusiasts of generative art alike.

II. RELATED WORK

A. Generative Adversarial Networks(GAN).

The development of image generation techniques has seen significant advancements with the advent of deep learning, particularly through the use of Generative Adversarial Networks (GANs). Introduced by Ian Goodfellow et al. in 2014, GANs have revolutionized the field of generative modeling by employing a two-player game between a generator and a discriminator, where the generator attempts to create realistic data, and the discriminator aims to distinguish between real and generated data. This adversarial training approach has proven highly effective in producing high-quality synthetic images that resemble real data distributions.

B. Conditional-Generative Adversarial Network

Building upon the traditional GAN framework, Conditional GANs (CGANs) introduced a conditional input to guide the generation process, leading to more controlled and targeted image synthesis. CGANs allow for the generation of specific content based on additional information such as class labels, textual descriptions, or even visual features from other images. This conditional approach has been widely used in applications such as image-to-image translation, style transfer, and character generation, where explicit control over the generated output is desired.

C. VGG16 Model

For feature extraction in this project, we utilized the pre-trained VGG16 model, which has been a standard choice in the field of computer vision due to its robust architecture

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and high performance in image classification tasks. Originally proposed by Simonyan and Zisserman, the VGG16 model has shown great potential in capturing intricate details in images through its deep convolutional layers. The effectiveness of VGG16 lies in its hierarchical feature representation, which extracts low-level features in the initial layers and progressively captures more abstract patterns in the deeper layers.

D. Cosine Similarity

Cosine similarity, a technique used to measure the similarity between feature vectors, has been employed in numerous computer vision tasks such as image retrieval and matching. This technique has been particularly useful for finding visually similar items in a dataset, as it effectively measures the angle between two vectors, indicating how closely related they are. In this project, cosine similarity played a crucial role in identifying the Pokémon that most closely matched the visual attributes of the human input image.

The application of deep feature extraction models like VGG16 for content generation and personalized image synthesis has been explored in various studies. For instance, works involving face-swapping, style transfer, and character generation have all benefited from the use of feature extraction techniques to condition the generation process [4]. These studies demonstrate how deep learning models can learn complex visual attributes that can be transferred or blended to create new forms of visual content.

While traditional GANs have been successful in generating realistic images, their application in character customization remains limited without the use of conditions. This project uses the power of CGANs to create personalized Pokémon by conditioning the generator on both human features and the most similar Pokémon. The integration of these methodologies—GANs, CGANs, VGG16 for feature extraction, and cosine similarity for matching—provides a decent pipeline for generating hybrid characters.

III. METHODOLOGY

The methodology of this project is divided into four key stages: data preprocessing, feature extraction using the VGG16 model, identifying the closest Pokémon using cosine similarity, and image generation using a Conditional GAN (CGAN). The pipeline aims to combine the unique attributes of human images with Pokémon features to create personalized hybrid characters.

A. Data Preprocessing

The data preprocessing phase involved preparing both the human and Pokémon image datasets for feature extraction and training.

Key steps included:

- **Image Conversion:** All images were converted to a consistent format (JPEG) to ensure uniformity across datasets.

- **Resizing:** Images were resized to dimensions of 64x64 pixels to match the input requirements of the feature extraction model and CGAN.
- **Normalization:** Pixel values were scaled to a range of [0, 1] for the VGG16 model and [-1, 1] for the CGAN to speed up training and improve stability.

B. Feature Extraction Using VGG16

To extract deep features from both human and Pokémon images, we used a pre-trained VGG16 model, specifically the fc1 layer. This layer was selected because it represents high-level features that capture detailed visual information useful for distinguishing between complex patterns in images. The steps involved in this phase were:

Feature Vector Generation: Images were passed through the VGG16 model up to the fc1 layer, and the resulting feature vectors (of size 25088) were extracted.

Saving Features: These feature vectors were saved into .npy files for both datasets to facilitate faster processing during Pokémon matching.

C. Pokémon Matching Using Cosine Similarity

To find the most similar Pokémon to a given human image, cosine similarity was employed. This step involved:

- **Feature Comparison:** The feature vector of the human image was compared to all Pokémon feature vectors using cosine similarity.
- **Closest Match Identification:** The Pokémon with the highest cosine similarity score (closest angle between feature vectors) was selected as the base Pokémon for hybrid character generation.

D. Conditional Generative Adversarial Network (CGAN) Architecture

The CGAN architecture was specifically designed to condition the image generation process on both the human features and the matched Pokémon. The architecture consists of two primary components: the generator and the discriminator.

Generator Architecture

The generator takes in a latent noise vector and a condition vector (human and Pokémon features) as input to produce a new image. Key layers and their roles include:

Dense Layer: The first dense layer transforms the concatenated input into a higher-dimensional space, creating the base structure of the generated image.

Reshape Layer: This layer reshapes the dense output into a small feature map of 8x8x256, forming the foundation for further upsampling.

Conv2DTranspose Layers: These layers perform upsampling using transposed convolutions, gradually increasing the resolution of the image while refining its features.

Batch Normalization: Batch normalization is applied after each transposed convolution to stabilize training and normalize the layer outputs.

Activation Functions: The generator uses ReLU activations for hidden layers to avoid gradient saturation and a tanh activation in the final layer to ensure the output pixel values are in the range $[-1, 1]$.

Discriminator Architecture

The discriminator is a binary classifier designed to distinguish between real and generated Pokémon images by examining both the image and the condition vector. Key components include:

Conv2D Layers: These layers extract hierarchical features from the input image through downsampling operations, capturing local patterns.

LeakyReLU Activation: LeakyReLU is used to prevent dead neurons and maintain a small gradient for negative values, aiding in stable training.

Flatten and Dense Layers: The image features are flattened and combined with the condition vector, allowing the network to learn more complex relationships.

Dropout Layer: Dropout is applied to reduce overfitting and increase the model's robustness.

Final Output Layer: The final dense layer uses a sigmoid activation to output a probability indicating whether the image is real or fake.

E. Training Process

The CGAN was trained using the following strategy:

Adversarial Training: The generator and discriminator were trained alternately in a two-player minimax game. The generator aims to fool the discriminator, while the discriminator aims to correctly classify real and generated images.

Loss Function: Binary cross-entropy loss was used to measure the performance of both models, with the generator seeking to minimize the discriminator's ability to detect fakes.

Optimizer: Adam optimizer was used for both the generator and discriminator, with tuned hyperparameters to ensure smooth convergence.

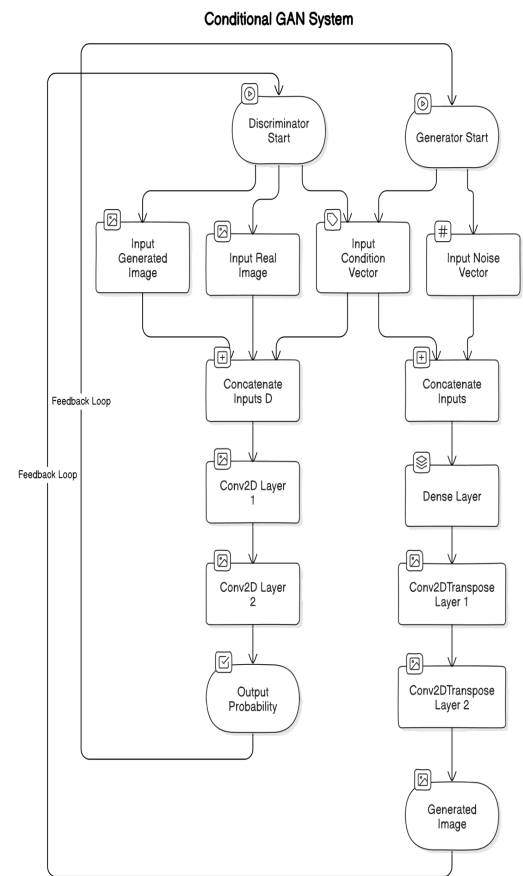
The use of a CGAN rather than a standard GAN allowed for the generation of personalized Pokémon characters by conditioning the generator on specific human attributes, resulting in more controlled and relevant hybrid images.

Tools Used:

- Jupyter Notebook
- Kaggle Notebook(30GB RAM, TPU)
- Param Shavak
- Tensorflow

IV. DIAGRAMS

High-level Architecture Diagram



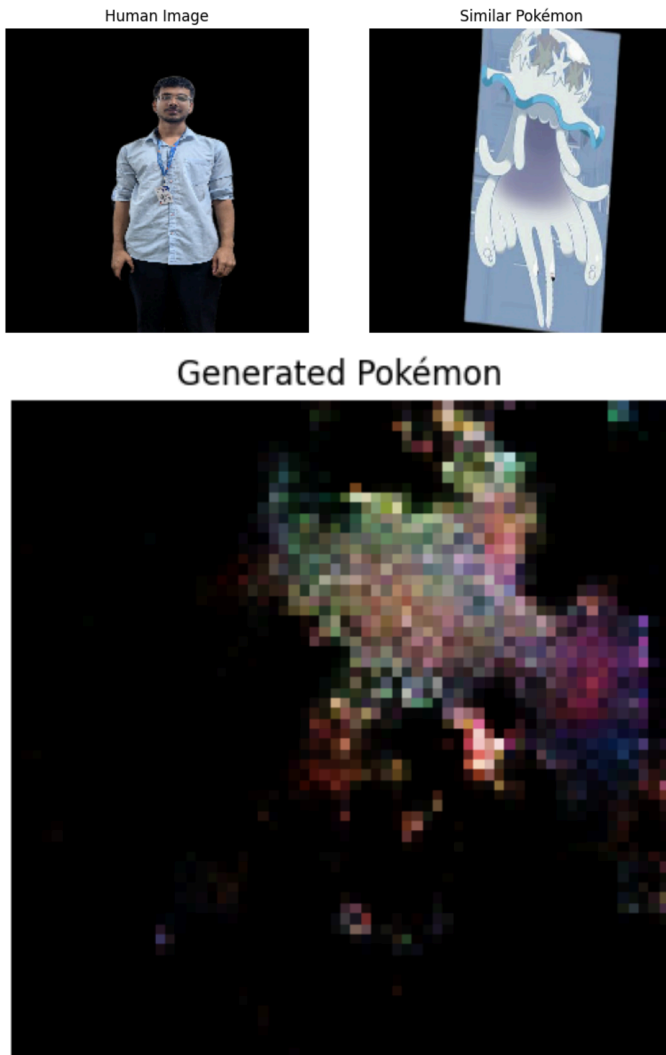
V. RESULTS

In this section, we present the outcomes of our Conditional GAN (CGAN) model, which was trained to generate personalized Pokémon images by blending human features with Pokémon attributes.

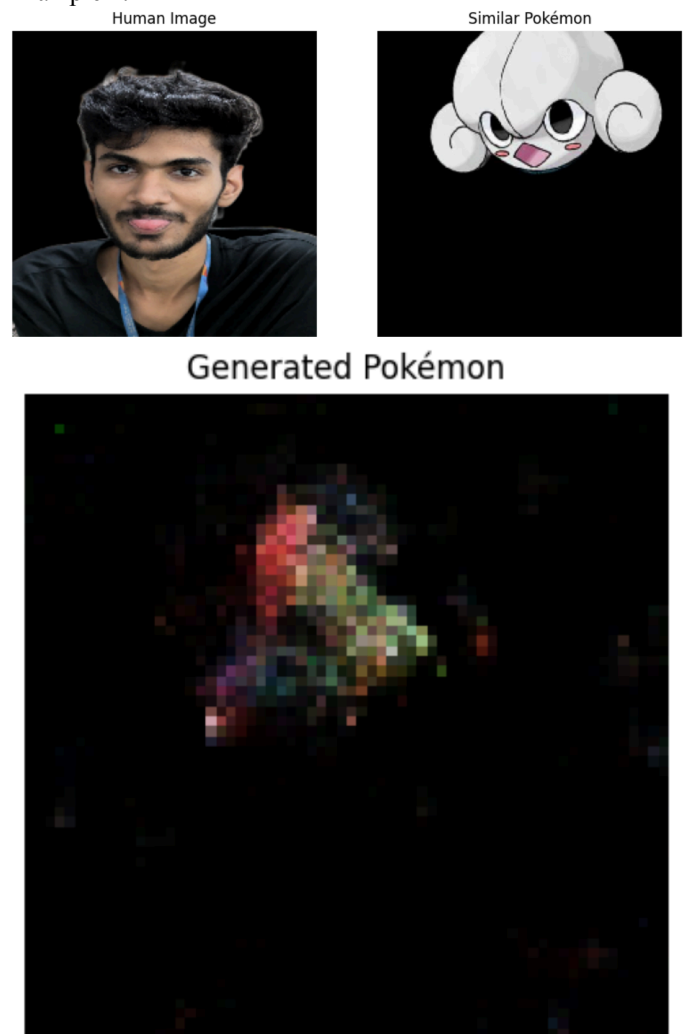
Figure Description:

- **Original Human Image:** The human image used as input for feature extraction.
- **Matched Pokémon Image:** The most similar Pokémon identified based on cosine similarity with the human image features.
- **Generated Pokémon Image:** The final output image created by the CGAN, illustrating the personalized Pokémon character with combined human and Pokémon features.

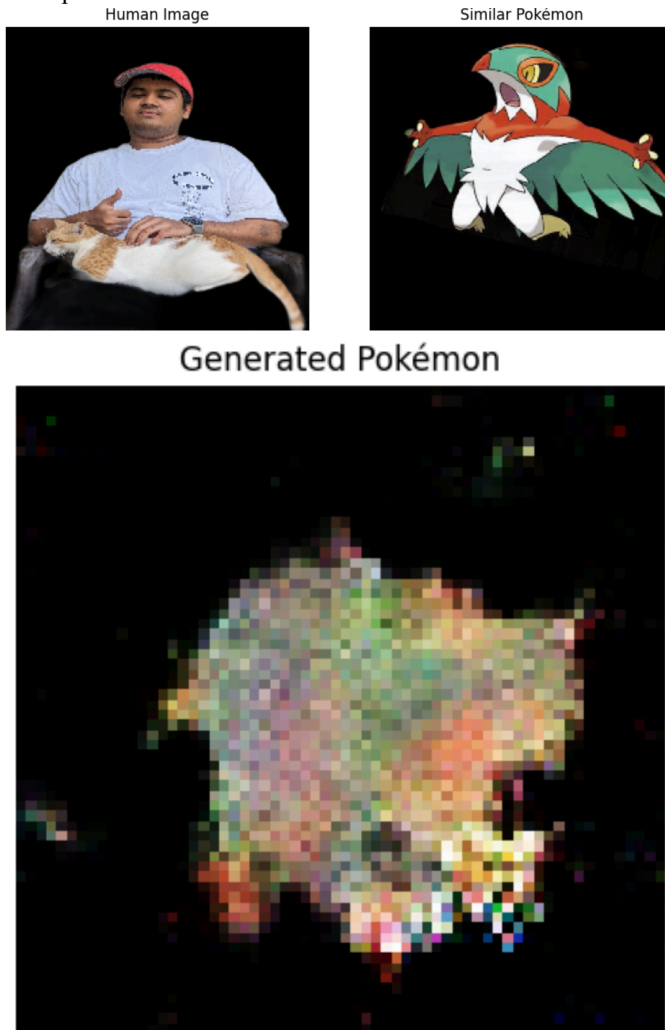
Example 1.



Example 2.



Example 3.



Analysis of Results

The results reveal several insights about the performance of our CGAN architecture:

- **Integration of Human Features:** The CGAN tries to integrate human-specific attributes into the Pokémon designs, as evident from some visible characteristics that resemble the human input like colour, size, boundary, etc.
- **Quality of Generated Images:** The use of ReLU activations in the generator allowed the model to produce the images even though not realistic, while the tanh activation ensured the pixel values remained within a range.

VI. CONCLUSION AND FUTURE WORK

In this project, we successfully demonstrated the integration of deep learning techniques with Generative Adversarial Networks (GANs) to generate personalized Pokémon characters using human visual features. By using a pre-trained VGG16 model for feature extraction and a Conditional GAN (CGAN) for image synthesis, we developed a system capable of blending human attributes with Pokémon designs to create unique and hybridized characters.

The feature extraction phase using the VGG16 model helped us to capture fine details of human images, such as facial features and textures, which were then used to identify the most visually similar Pokémon through cosine similarity. The Conditional GAN then utilized this information to generate new Pokémon images conditioned on the extracted human features, leading to visually coherent and creative results. The generator and discriminator architectures were designed to ensure outputs, with careful use of activation functions and layers to stabilize training. The results indicate that our CGAN-based approach is effective in generating images but requires more data to better reflect the fusion of human and Pokémon characteristics.

Future Work

While the results of our project are not that sharp, there are several areas for potential improvement and future exploration:

- **Enhanced Feature Extraction:** Experimenting with more advanced deep learning models, such as ResNet or Inception, could improve the quality and specificity of feature extraction.
- **Data Augmentation:** Further augmenting the human and Pokémon datasets could help the model generalize better and handle a wider variety of input images.
- **Diverse Generative Models:** Exploring alternative generative models like StyleGAN or Diffusion Models may provide even more realistic and high-resolution character outputs.
- **Real-time Applications:** Developing a real-time interface for users to upload their images and instantly receive customized Pokémon could significantly enhance the user experience.

In conclusion, the project highlights the potential of using GANs and deep feature extraction for personalized content generation. It bridges the gap between artificial creativity and user-specific customization, paving the way for more personalized experiences in gaming, digital art, and beyond.

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